A HANDOUT ON KNOTWORK AND THE SPECTRE

The Spectre Knot

A single-strand Celtic knot on an aperiodic monotile

One endless ribbon through a tiling that never repeats

Justin Roughley • 7 Oct 2026

A tile that never repeats. Squares and hexagons tile the plane periodically: slide the pattern and it lands on itself. Penrose found pairs of tiles that can only tile non-periodically; whether one shape could force this on its own stayed open for decades, as the *einstein* problem ('one stone'). In 2023 David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss solved it, first with the *hat*, which needs some copies flipped over, then weeks later with the **spectre**, which needs none. The tile used here is the straight-edged form of the spectre, their fourteen-sided polygon Tile(1,1). Laid without flipping, its copies tile exactly as spectres do, and only in patterns that never repeat. Curving its edges gives the strict spectre, which rules out flipped copies by its shape alone. The knot is the same on both.

A knot that never repeats. A Celtic knot is one ribbon woven over and under itself, usually built by repeating a single cell. Here the cell is the spectre. Every tile carries the same small drawing: fourteen *partial strands*, pieces of ribbon from one edge of the tile to another, which join up where tiles meet (page 2). The question is whether one drawing can make every piece on every tile of an infinite tiling join into **a single strand**. On any repeating tiling that is impossible. On the spectre it can be done. However the tiles cover the plane, no strand ever closes into a loop (the *Endless Strand Theorem*); and in the *nest tiling* drawn here, grown from one tile (foot of the page), it is one endless ribbon through every tile (the *Ariadne Theorem*).

1 Why it is one strand

Spectres group into supertiles of eight or nine, these into larger ones, and so on for ever. Seen from outside, every supertile looks the same: the ribbon hooks round each corner of its outline, except at the two ends of one edge, where two ribbons cross. Where supertiles meet along a run of edges the ribbons zigzag along it and only the run's parity matters, so each level is built from the one below by a walk on a small map. The pattern passes up the levels like falling dominoes, no ring can form, and pieces carried up a nested chain of supertiles all end on one ribbon.

The computer checked it all up to level 9 (133 million tiles). Beyond that, the spectre paper proves every level is built by the same rule (its Theorem 5.1), so edge lengths obey one linear rule, their parities never change, and the small map repeats every two levels for ever.

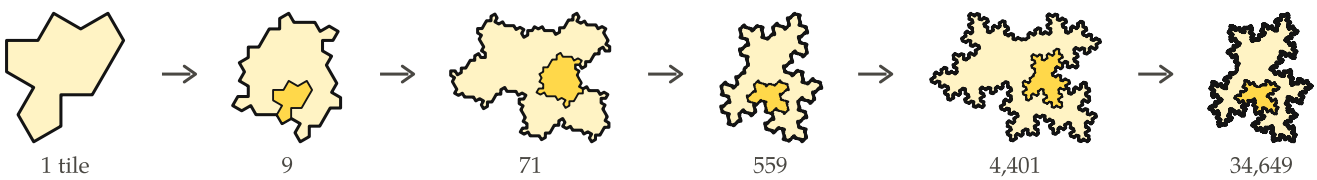
2 Watching it join up

The number of strands that a nested block's ribbon pieces lie on, as the block around it grows:

block	*pieces*	*1 up*	*2 up*	*3 up*	*4 up*
1 tile	14	5	2	**1**	**1**
9 tiles	46	10	2	2	**1**
71 tiles	182	37	2	**1**	**1**
559 tiles	758	146	2	2	**1**
4,401 tiles	3,198	613	2	**1**	**1**

The large first numbers are pieces on the stretch of a block's edge shared with the next block; they join a level later. Odd-level blocks need four levels, not three: one strand ends on edge shared by the next two blocks up, and waits for a block that encloses it (pages 5 to 7).

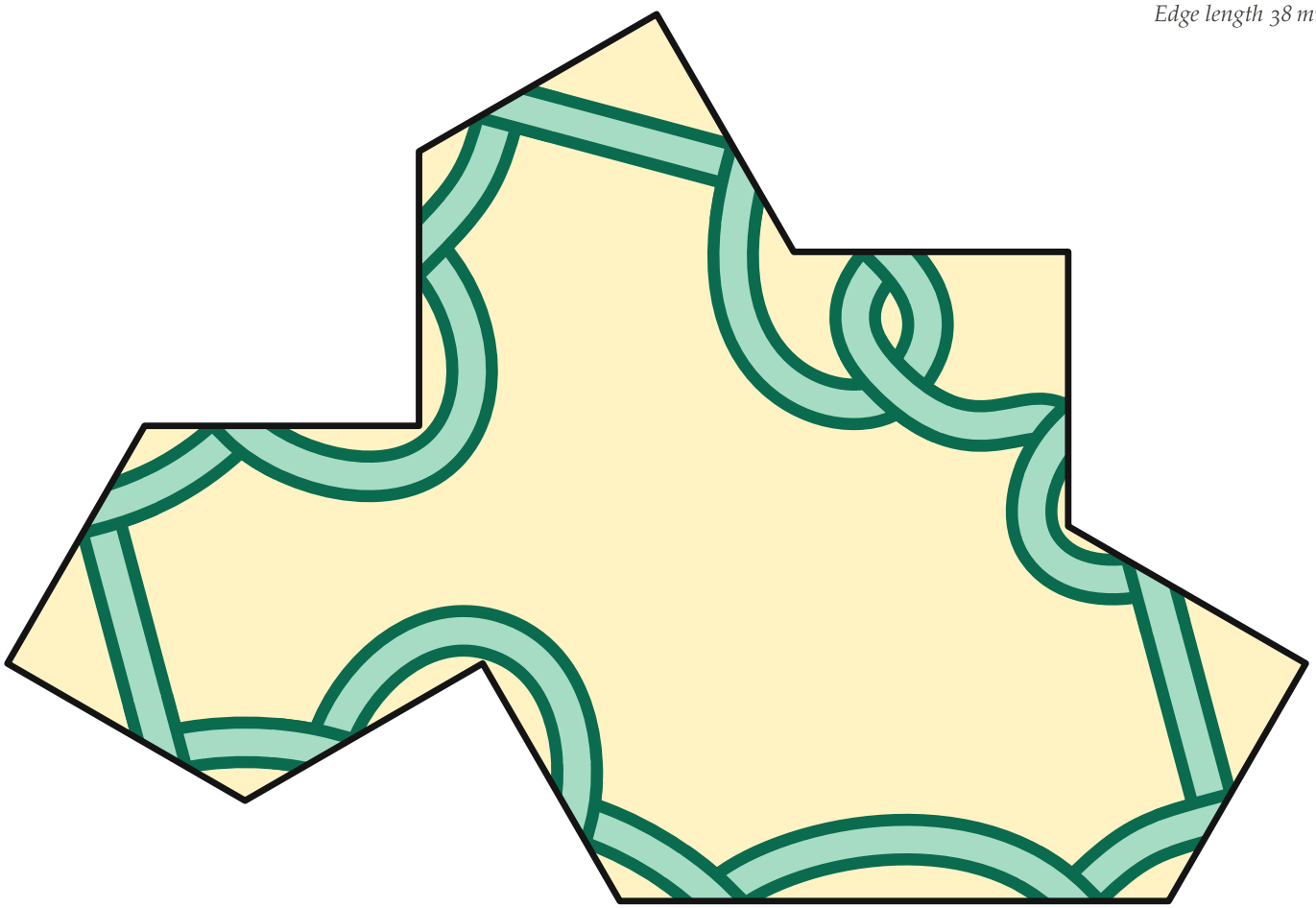
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss, *A chiral aperiodic monotile*, Combinatorial Theory 4(2) (2024), arXiv:2305.17743; and *An aperiodic monotile* (the hat), Combinatorial Theory 4(1) (2024), arXiv:2303.10798. Knot design 0.1:01. Made with AI; how, and what is checked: cortado-j.github.io/SPECTRE_KNOT/made.html



Each block is the shaded part of the next, all the same way up. The chain fills the plane: it is the nest tiling drawn here.

The Spectre Knot · one tile

Edge length 38 mm



One crossing inside

Two ribbons cross at every edge of the tile, at a point shared with the neighbour across it, and they cross just once inside the tile. Fourteen pieces of ribbon join the edges up: twelve short hooks that turn round a corner, and two longer pieces that leave the same edge and cross in the middle.

Off-centre, for looks

Each crossing sits 0.15 of an edge length off the midpoint, alternately one way and the other round the tile. Spectres always meet even-numbered edge to odd-numbered edge, so neighbours' crossings land on the same point. The shift only makes the curves more pleasing: at the midpoints the ribbons would join up just the same.

Two theorems

However the tiles cover the plane, no strand ever closes into a loop: the *Endless Strand Theorem*. In the nest tiling of page 1 it is one strand: the *Ariadne Theorem*. Computer checks suggest some other tilings have three.

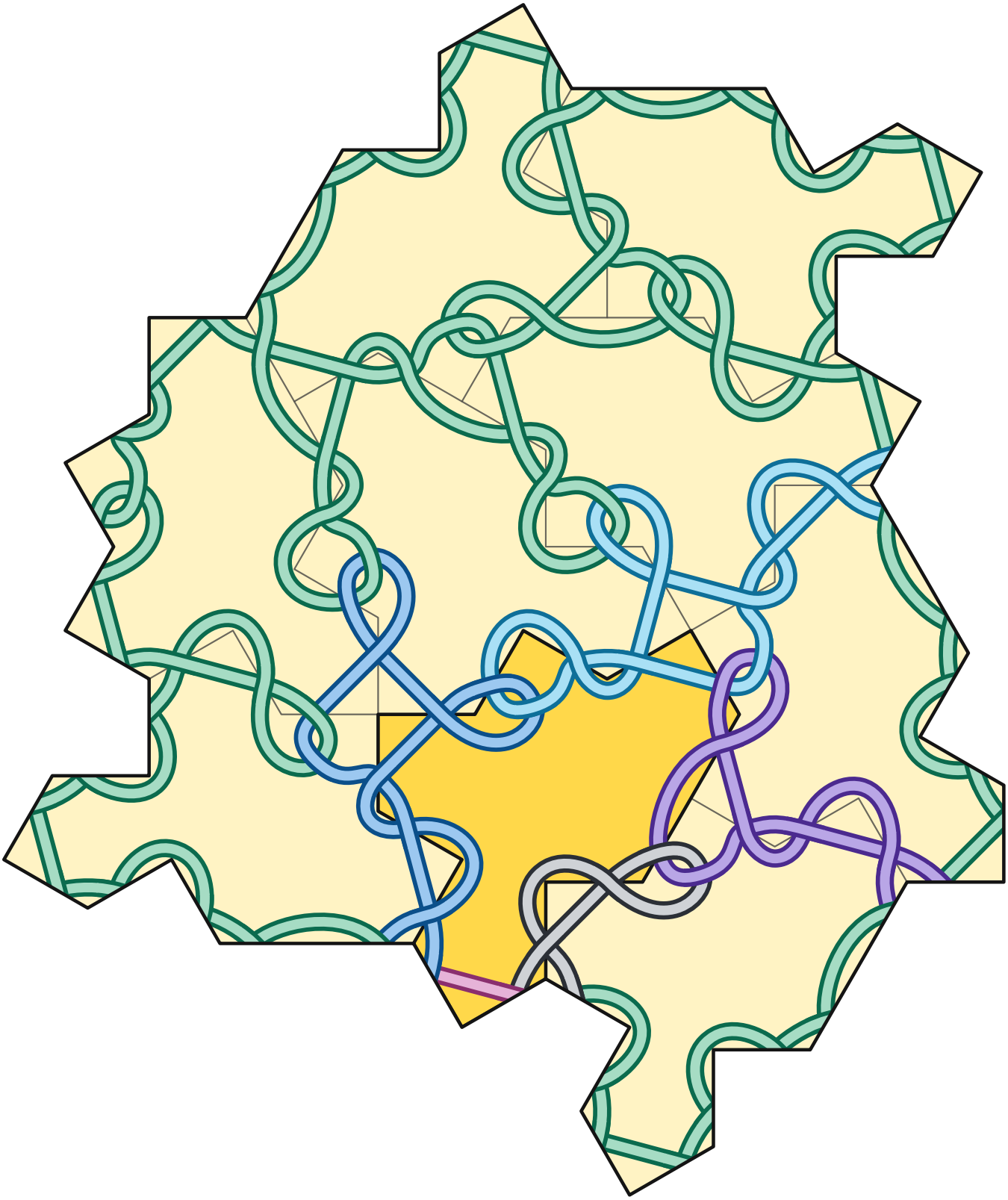
One tile, one strand

Copy this tile across the nested tiling of page 1 and the pieces join into a single ribbon that visits every tile, never closes up and never repeats. The over-and-under alternates all the way, and one printed tile serves everywhere. The traced tile on the following pages is on one strand by the 559-tile supertile.

Things to try

1. Make a few tiles from page 8 and copy a patch from pages 3 to 7. Follow any ribbon with a finger: it always leaves the patch. Why must it?
2. Show that no design repeated on every tile of a periodic tiling, such as squares or hexagons, can ever give a single strand. Hint: what does a translation do to the strand?

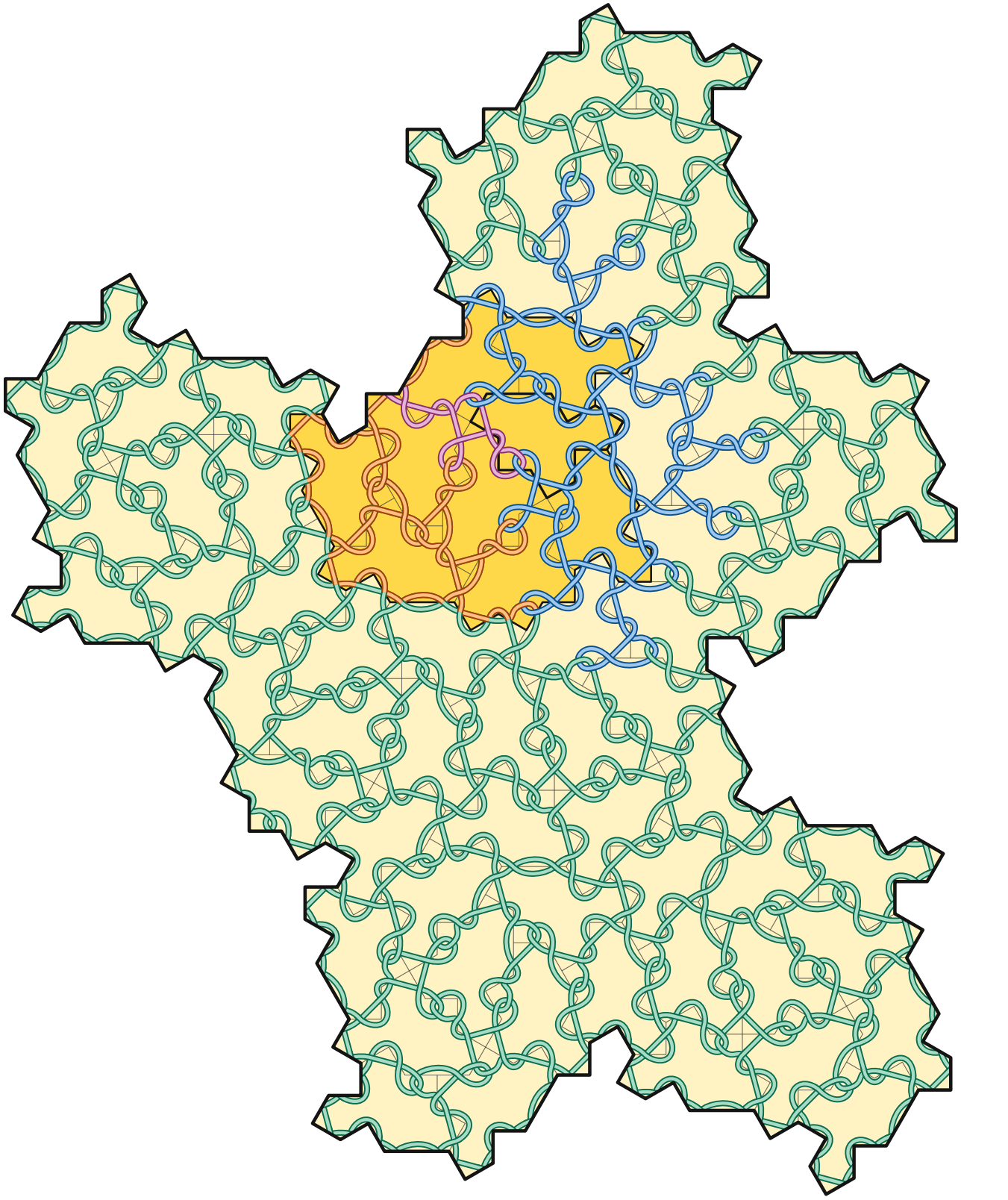
Level 1 supertile · 9 tiles



a spectre tile
the previous page's block
the ribbon
its ribbon pieces
strands through the original tile
the original tile

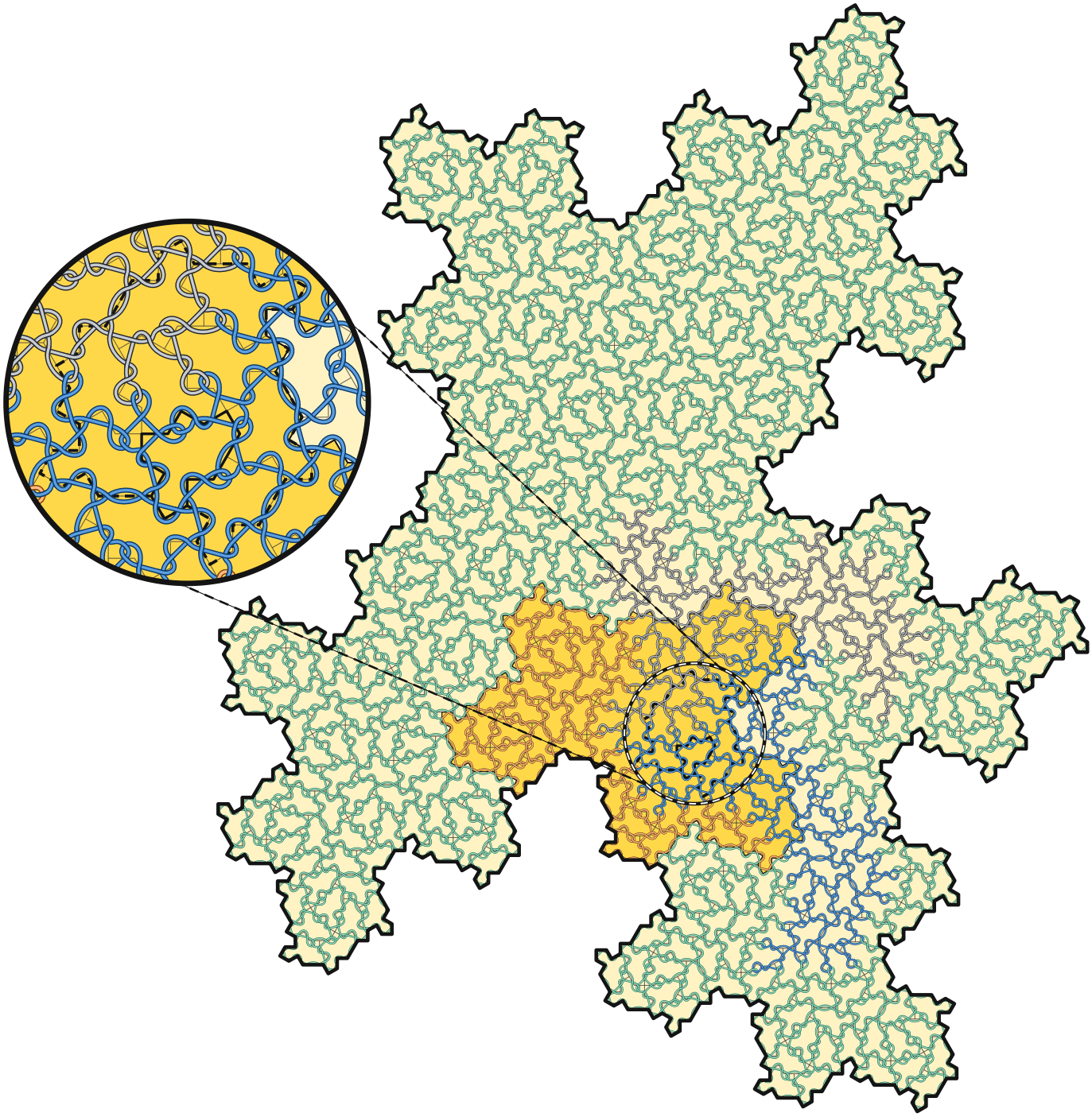
Outlined in black and shaded brighter yellow: the original tile from page 2. Its 14 ribbon pieces lie on **5 different strands** of this supertile, each drawn in its own colour.

Level 2 supertile · 71 tiles



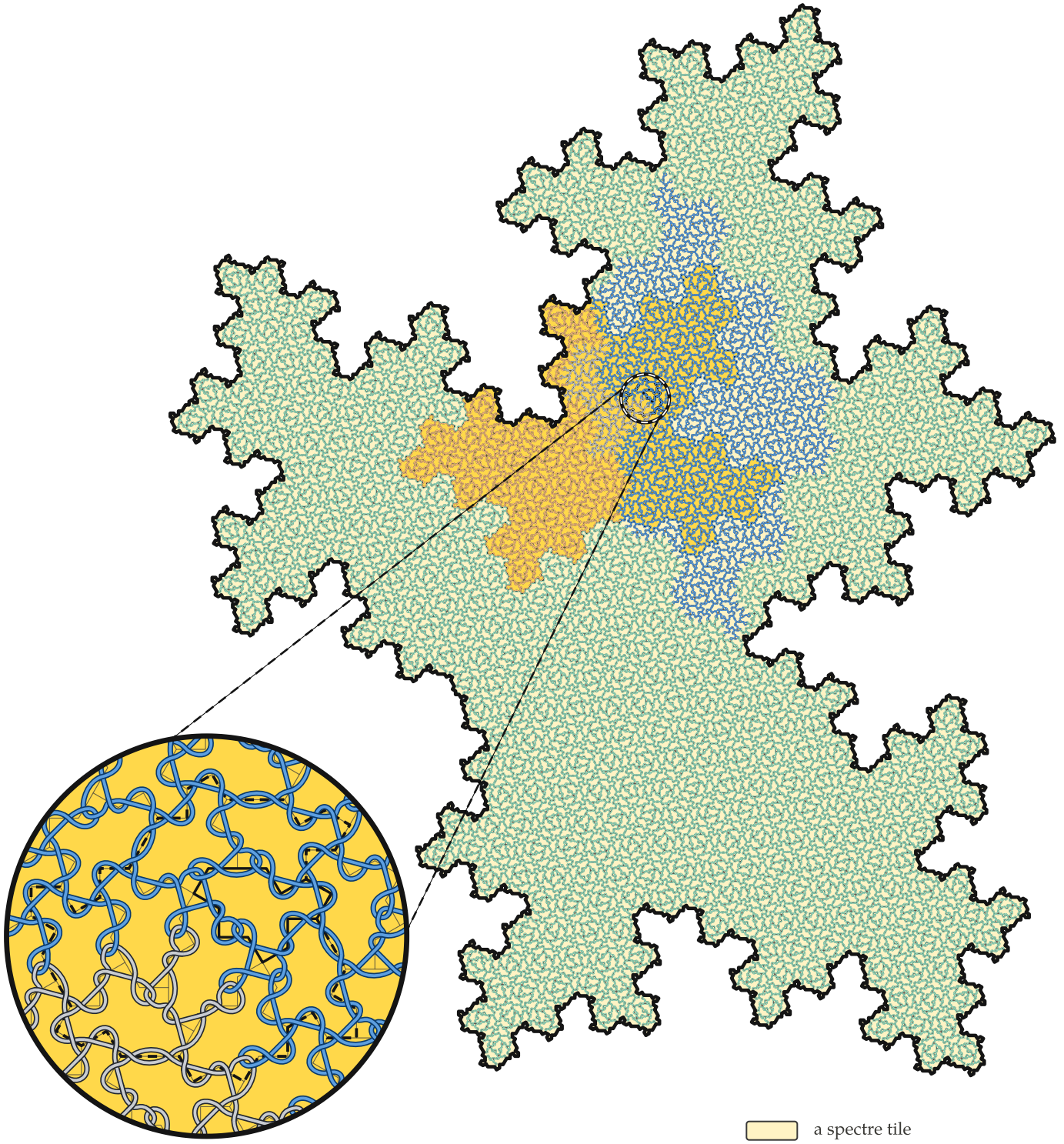
Brighter yellow: the 9-tile supertile of page 3; its 46 ribbon pieces (orange) lie on 10 strands here. **Outlined in black:** the original tile. Its pieces now lie on just **2 strands**, blue and purple.

Level 3 supertile · 559 tiles



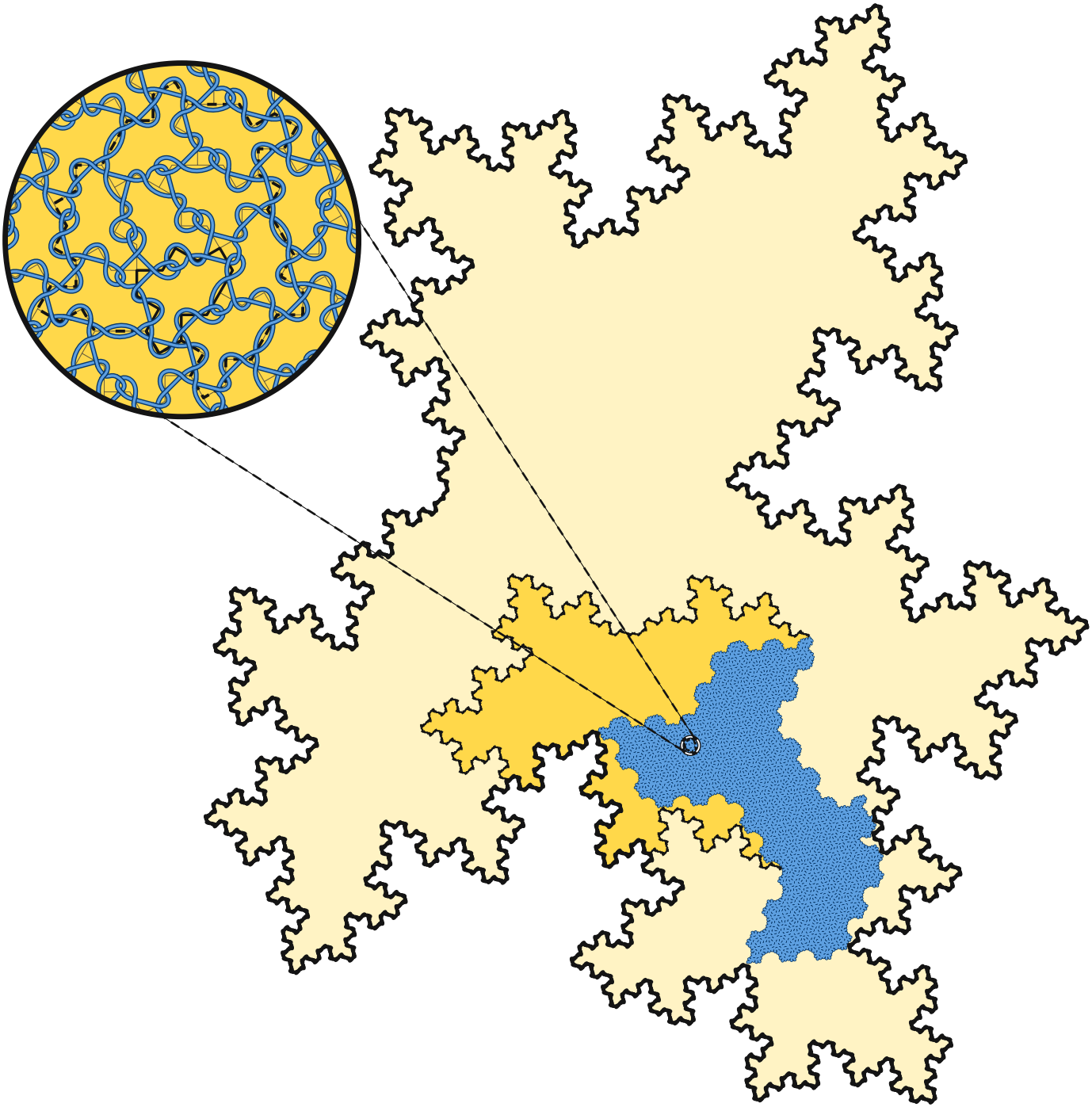
Brighter yellow: the 71-tile supertile of page 4; its 182 pieces (orange) lie on 37 strands here. **Circled, dashed:** the 9-tile supertile of page 3, with the original tile inside it. The original tile is now on **one strand** (blue, 68 tiles), but the 9-tile block is not: the rest of its pieces lie on a **second strand** (charcoal, 69 tiles). **Inset:** the circle, magnified 3×.

Level 4 supertile · 4,401 tiles



Brighter yellow: the 559-tile supertile of page 5 (758 pieces on 146 strands). The 9-tile block is **still on two strands**. The blue one has grown to 687 tiles; the charcoal one is exactly as before, 69 tiles, because both its ends sit on a stretch of edge that this supertile shares with the 559-tile block. It can only grow once a bigger block encloses that stretch. **Inset:** the circle, magnified 8×.

Level 5 supertile · 34,649 tiles



Brighter yellow: the 4,401-tile supertile of page 6. Here at last the two strands meet: all 46 pieces of the 9-tile block, and with them the original tile, lie on **one strand**, which winds through 3,035 tiles (blue). This is four levels above the 9-tile block, which is why the proof climbs four levels: blocks at odd levels need four, blocks at even levels, like the original tile, need three. At this size only that strand is drawn. **Inset:** the circle, magnified 20×, with every ribbon drawn: by now every one of them belongs to that single strand.

- a spectre tile
- the previous page's block
- strands through the original tile
- the 9-tile supertile
- the original tile

The Spectre Knot

One tile, front and back

Front: the knot tile. Back: what it is. Edge length 41 mm.



The Spectre Knot

A single-strand Celtic knot on an aperiodic monotile

The tile is the straight-edged form of the spectre, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss: a single shape whose copies cover the whole plane, but only in patterns that never repeat.

The knot, the Spectre Knot, is the ribbon on the front. Justin Roughley came up with the original idea of looking for a mono-stranded aperiodic knot, and then used AI to search for it (2026). It is the same on every tile, yet across the nest tiling, grown outwards from one tile, its fourteen pieces join into **one endless strand:** a single Celtic knot through every tile, with no closed loops and no second strand, weaving over and under all the way, and never repeating.

No repeating tiling can do this: its symmetries would have to slide the one strand along itself in two independent directions at once, which a single line cannot do. It takes a tiling that never repeats, and the design is about as simple as it can be, with just one crossing inside the tile.

Why it works: the tiles group into supertiles of 9, 71, 559, 4,401 tiles and on for ever, and each larger supertile gathers the ribbon pieces onto fewer strands. Exact checks of the smallest cases, with the spectre paper's theorem that every level is built by the same rule, show that every piece ends up on the same one; a computer also followed it up to 133 million tiles. In every spectre tiling no strand ever closes into a loop (the Endless Strand Theorem); one strand (the Ariadne Theorem) is proved for the nest tiling only, and computer checks suggest some others have three. The proof was made with AI and has not been checked by a mathematician.

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1. Cut round the outer black line, as one piece.
2. Fold along the dashed line, printed sides out.
3. Glue the halves back to back.

Make several and lay them edge to edge: the ribbons meet where the edges touch and cross there at right angles.